



UNIVERSITY OF ENGINEERING AND TECHNOLOGY MARDAN

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SECTION	A
DEPPT	COMPUTER SCIENCE
SUBJECT	DLD LAB
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LAB TASK : 1

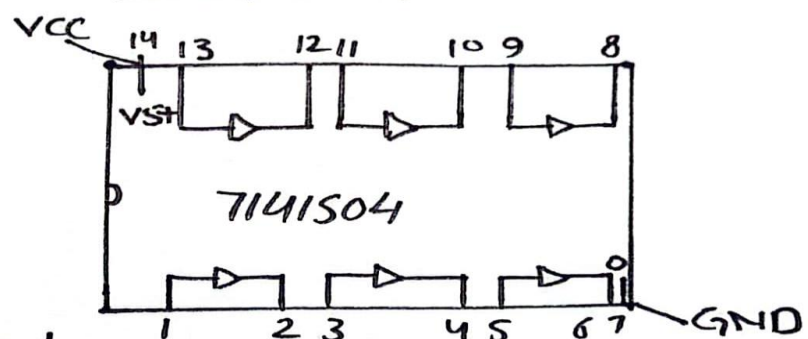
Not Gate IC :- It is a single input & output digital device or logic gate.

Explanation :- A not gate is also known as inverter: is a fundamental logic gate in digital electronics. It has one input and one output.

Truth Table :-

Input	Output
0	1
1	0

Diagram



Trainer Board

A trainer board is a teaching tool used in electronics education, & featuring components like switches, ICs and gates is used.

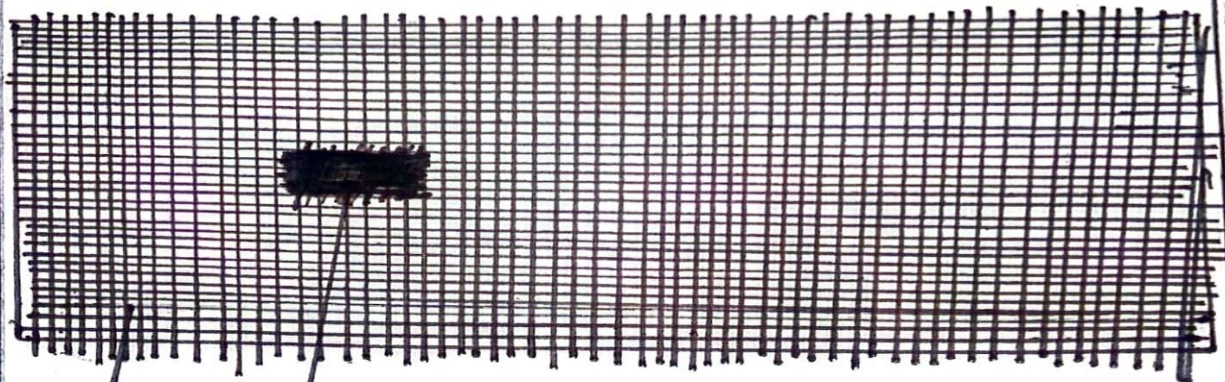
Explanation

Digital logic Design a trainer board referred to a specialized educational tool used to teach components related to digital logic circuit. These boards typically include components such as logic gates (And, OR, Not)

multiplexers, decoders in incooding of blocks.

Components:- Main components use in trainer board is DLD.

- 1) Board It is main component in trainer board connects the not gate and different IC's in trainer board.

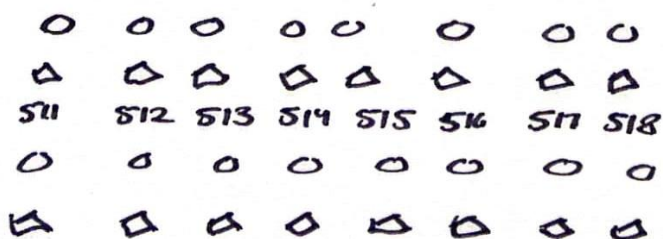


board

Not Gate IC, attached.

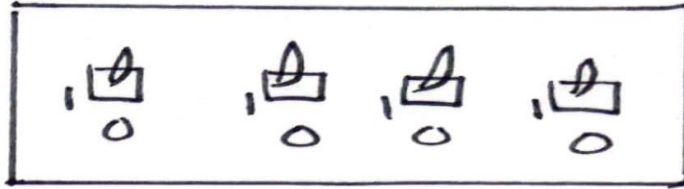
- 2) State Monitor :- (output)

It is used for IC gate No IC, for output attach through wise output of IC to state monitor of machine.



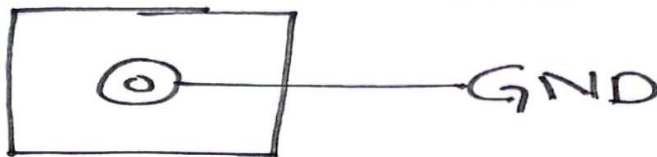
Logic Switches (input):- Logic switches is used for input of IC to trainer board it is part of trainer board

which receive input.



Diagram

GND (Ground Earth) It is a part of trainer board which is used for earth of current it observed more electricity of signal attach 7 points of IC is GND point attached with GND.



Power (+5v) voltage:- This circuit is used for power +5v which show the power of ICs attached with trainer through wire.

Steps of Lab Work.

Step 1:- The IC of not gate is attached in main board of trainer machine in place it fixed in IC and trainer machine.

Step 2:- 2nd I need three wire for fixed the trainer machine to ICs and work and experiment.

Step 3 :- We attached the first whole of 1st point (P_1) if IC is input to input logical switches through the wired and connect.

Step 4 :-

I 2nd whole second point (P_2) and output of state monitor is attached through wire.

Step 5 :-

We have the 7th point (P_7) is GND used for ground in IC, is attached combine the fixed supply DC and GND through wire.

Step 6 :-

We have the 14 points of (P_{14}) of IC is attached Fix power supply in power (+5V) through wire.

Main Work

I have ready of all steps of experiments, check it.

① when Input is (0) output is (1).

② when Input is (1) output is (0).

Application of Not Gate.

Digital Logic:-

In digital circuit not gate for single inversion. They are often used in combinations with others gate (And, OR, Not).

Memory Devices

Not gate are essential in memory devices like - flops and latches where signal inversion is required for storing and retrieving data.

Clock Circuit:

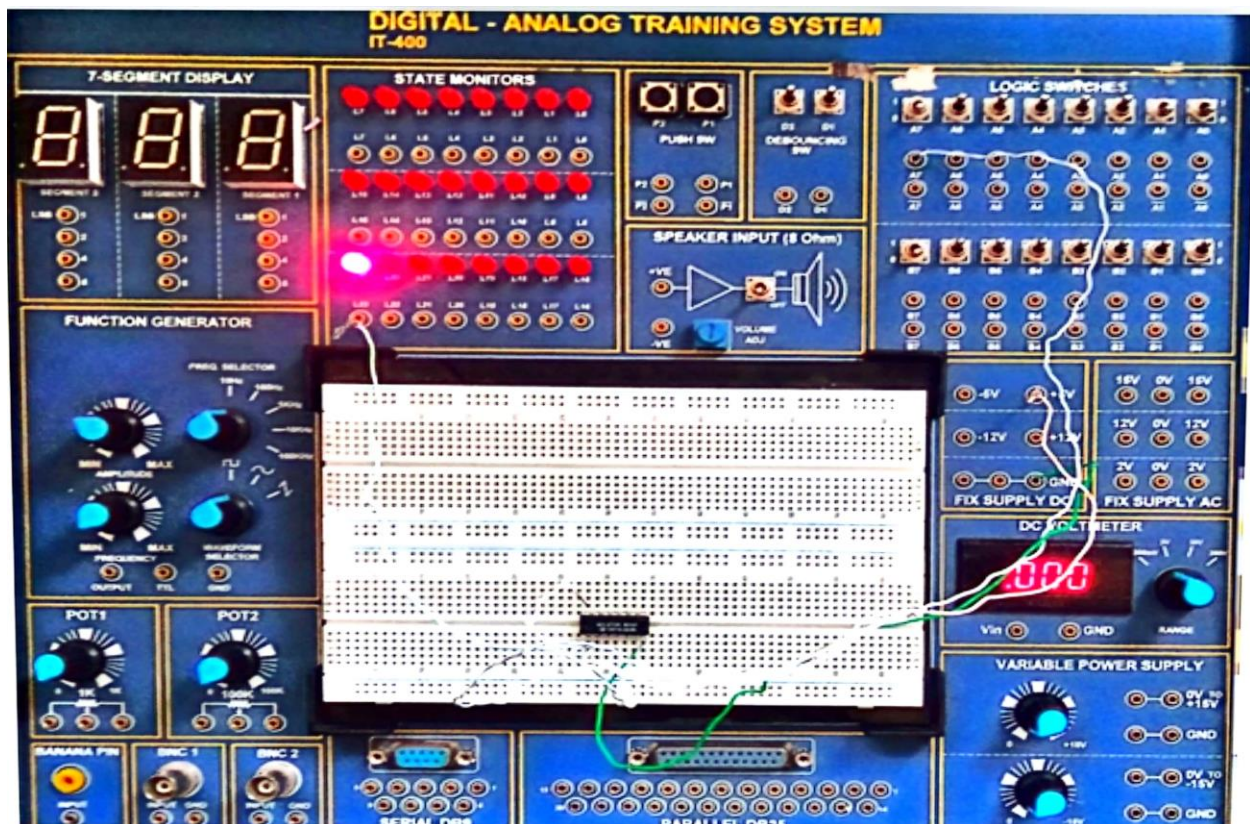
In clock circuits not gates are used to generate complementary clock signals. This is circuit in synchronizing operations within digital systems.

Signal Processing

Not gates are utilized in signal processing circuits where signal inversion is necessary for specific operation like phase.

LED Drivers:

In LED display systems, not gates can be employed to drive segments of display by providing inverted signals.



[illegible][illegible]

